Control Hazard

- This type of dependency occurs during the transfer of control instructions such as BRANCH, CALL, JMP, etc.
- On many instruction architectures, the processor will not know the target address of these instructions when it needs to insert the new instruction into the pipeline.
- Due to this, unwanted instructions are fed to the pipeline.

Example

- Consider the following sequence of instructions in the program:
  100: I_1
  101: I_2 (JMP 250)
  102: I_3
  .
  .
  250: BI_1
- Expected output: I_1 -> I_2 -> BI_1
- Output Sequence: I_1 -> I_2 -> I_3 -> BI_1

Control Hazard

INSTRUCTION/ CYCLE	1	2	3	4	5	6
I_1	IF	ID	EX	MEM	WB	
I_2		IF	ID (PC:250)	EX	Mem	WB
I_3			IF	ID	EX	Mem
BI_1				IF	ID	EX

Resolve the problem

INSTRUCTION/ CYCLE	1	2	3	4	5	6
I_1	IF	ID	EX	MEM	WB	
I_2		IF	ID (PC:250)	EX	Mem	WB
Delay	–	–	–	–	–	–
BI_1				IF	ID	EX

Solution for Control dependency

- Branch Prediction is the method through which stalls due to control dependency can be eliminated. In this at 1st stage prediction is done about which branch will be taken.
- For branch prediction Branch penalty is zero.
- **Branch penalty :** The number of stalls introduced during the branch operations in the pipelined processor is known as branch penalty.